

CYBERSHIELD NEXUS

Cybercrime Assistance & Intelligence Platform

Complete Technical & Product Scope Document

Version 1.0 | Confidential | 2025

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1. Project Scope

1.1 Problem Statement

Cybercrime is one of the fastest-growing categories of crime globally. Victims — whether individuals, small businesses, or enterprises — consistently encounter three barriers when trying to respond: they do not know how to report incidents correctly, they lack access to expert guidance on immediate containment steps, and authorities struggle to identify emerging trends buried in isolated reports. CyberShield Nexus eliminates these barriers by providing a single, intelligent platform that guides victims through every stage — from incident identification to evidence submission, authority reporting, legal escalation, and preventive hardening.

1.2 Target Users

User Segment	Why They Use CyberShield Nexus
Individuals / Citizens	Victims of phishing, identity theft, online fraud, stalking, or financial scams need guided, jargon-free support.
Small & Medium Businesses	Organizations lack in-house security teams and need affordable, intelligent guidance after breaches or ransomware.
Enterprises & CISOs	Large organizations need threat intelligence, risk scoring, and coordinated incident management workflows.
Cybercrime Officers	Law enforcement needs structured reports, aggregated heatmaps, and case prioritization tools.
Security Analysts	SOC analysts need cross-case threat correlation, emerging pattern alerts, and exportable threat intelligence.
System Administrators	Platform admins need user management, system health monitoring, and audit controls.

1.3 Unique Value Proposition

- AI-driven personalized guidance — not just a static FAQ or a generic form.
- End-to-end lifecycle support — from first report to authority escalation and case closure.
- Aggregated risk intelligence — individual reports feed a collective threat map that benefits all users.
- Evidence-first architecture — built specifically for handling sensitive digital evidence securely.
- Role-aware access control — different views and capabilities for public users, officers, analysts, and admins.

1.4 Real-World Use Cases

Use Case 1 — Phishing Attack Victim

Scenario: A user receives a suspicious email, clicks a link, and realizes credentials were stolen.

Flow: They file an incident report, upload a screenshot of the email, and the AI classifies it as credential phishing.

Output: Step-by-step containment advice (change passwords, enable MFA, contact bank), a pre-filled police report draft, and an alert to other users in the same region.

Use Case 2 — Business Ransomware Incident

Scenario: A small business has all files encrypted; attacker demands Bitcoin payment.

Flow: The IT manager files an incident, uploads ransom note and sample encrypted file. AI classifies threat type, suggests isolation steps.

Output: Priority escalation to cybercrime unit, legal guidance on paying/not paying, decryption resource links, and a risk score update for the sector.

Use Case 3 — Authority Trend Analysis

Scenario: A state cybercrime cell wants to understand whether investment scams are rising in a district.

Flow: Officer opens the Analytics Dashboard, filters by region and crime type.

Output: Heatmap highlighting hotspot areas, monthly trend chart, case list for investigation, and automated risk alert triggered for nearby districts.

2. User Roles & Permissions

CyberShield Nexus implements Role-Based Access Control (RBAC) across four primary roles. Each role has a distinct permission set aligned to their real-world responsibilities and data sensitivity requirements.

2.1 Citizen / Public User

The primary consumer-facing role. Any individual or organizational representative can register as a public user without prior approval.

Capability	Actions	Data Access
Incident Reporting	Create, edit, submit cybercrime reports	Own reports only
Evidence Upload	Upload screenshots, documents, videos	Own evidence files only
Case Tracking	View status updates on submitted cases	Own cases only
AI Guidance	Receive personalized security recommendations	Own guidance history
Knowledge Center	Browse educational articles and guides	All public content
Alerts	Receive threat alerts relevant to their region/type	Region-scoped alerts only

2.2 Cybercrime Officer

Verified law enforcement or regulatory personnel with authority to review, investigate, and escalate reports. Requires admin approval and credential verification before activation.

Capability	Actions	Data Access
Case Review Queue	View, assign, and update case statuses	All reports in their jurisdiction
Evidence Access	Download and examine uploaded evidence	All evidence linked to assigned cases
Case Escalation	Mark cases for inter-agency escalation	All cases in their jurisdiction
Threat Heatmap	View geographic distribution of incidents	Region-scoped heatmap
Report Export	Export PDF/CSV reports for legal proceedings	Their jurisdiction's data
Bulk Alerts	Issue public threat advisories	Alert management only

2.3 Security Analyst

Cybersecurity professionals analyzing threat patterns, correlating incidents, and producing intelligence feeds. May be internal (platform-employed) or external (enterprise-licensed).

Capability	Actions	Data Access
Threat Intelligence	Correlate incidents, tag threat actors	Anonymized aggregate data
Risk Prediction	Run and schedule risk scoring models	All risk model outputs
Pattern Analysis	Cluster similar attack vectors	Cross-case metadata (anonymized)
Alert Management	Create and publish threat alerts	Full alert CRUD
Analytics Dashboard	Full access to all charts and metrics	Platform-wide aggregate data
API Access	Export threat feeds via REST API	Scoped to licensed tier

2.4 System Administrator

Platform-level administrators responsible for system configuration, user management, audit trails, and infrastructure health. Full access is restricted to a maximum of two personnel with hardware-token MFA.

Capability	Actions	Data Access
User Management	Create, approve, suspend, delete accounts	All user records
Role Assignment	Assign and revoke roles	All roles and permissions
Audit Logs	View complete activity audit logs	Full audit trail
System Configuration	Configure AI model parameters, alerts	All system settings
Data Retention	Set and enforce data retention policies	All data tiers
Infrastructure Health	Monitor uptime, error rates, storage	All system metrics

3. Core Features

3.1 Incident Reporting System

Attribute	Details
What it does	Provides a structured, guided form for capturing cybercrime incident details. Uses progressive disclosure — questions adapt based on earlier answers (e.g., phishing flows differ from ransomware flows).
Why it is useful	Structured data enables accurate AI classification, prevents incomplete reports, and ensures authorities receive actionable, consistently formatted information.
Data required	Incident date/time, description, crime type (auto-suggested), victim details, suspect info (optional), affected assets, financial loss, prior authority contact.
Expected outputs	A structured JSON incident record, a human-readable case summary PDF, and an AI threat classification with confidence score.

3.2 Evidence Upload System

Attribute	Details
What it does	Secure multi-file upload supporting images (PNG/JPG), videos (MP4/MOV), documents (PDF/DOCX/XLSX), email exports (.eml), and log files (.txt/.csv). Files are virus-scanned, metadata-stripped, and encrypted at rest.
Why it is useful	Digital evidence is admissible only when chain of custody is maintained. The system timestamps every upload, records uploader identity, and generates SHA-256 hashes for integrity verification.
Data required	File binary, file type, associated case ID, uploader ID, upload timestamp, optional description/annotation.
Expected outputs	Encrypted file stored in S3-compatible storage, hash record in database, evidence card displayed in case view, AI-extracted metadata (OCR text from images, email headers from .eml files).

3.3 Case Tracking

Attribute	Details
What it does	Real-time status tracking through a defined case lifecycle: Submitted > Under Review > AI Analysis Complete > Assigned to Officer > Investigation In Progress > Resolved / Closed.

Why it is useful	Victims often feel helpless after reporting. Transparent status tracking maintains trust and enables follow-up actions.
Data required	Case ID, current status, assigned officer (if applicable), status history log with timestamps, automated/manual update notes.
Expected outputs	Visual status timeline, email/SMS notifications on status changes, estimated resolution time based on case type.

3.4 Cyber Threat Identification Engine

Attribute	Details
What it does	NLP-based classification engine that analyzes incident text and evidence metadata to assign a threat category, sub-type, and severity score.
Why it is useful	Eliminates manual triage by officers, enables consistent categorization, and routes cases to the correct specialist queue automatically.
Data required	Incident description text, evidence metadata, prior case history (for repeat offenders), financial loss amount, affected asset type.
Expected outputs	Primary crime type (e.g., Phishing), sub-type (Spear Phishing / Business Email Compromise), CVSS-inspired severity score 1-10, confidence percentage, flagged keywords.

3.5 Personalized Guidance Engine

Attribute	Details
What it does	Generates a prioritized, role-appropriate action plan for each incident. Actions are sequenced (immediate > short-term > long-term) and linked to external resources.
Why it is useful	Most victims do not know what to do after a cyberattack. Guided, step-by-step advice reduces further harm and improves evidence preservation.
Data required	Classified crime type, victim profile (individual/business), affected platforms, financial impact, jurisdiction.
Expected outputs	Numbered action checklist (e.g., 1. Disconnect device 2. Change passwords 3. Contact bank), links to official reporting portals, legal guidance summary, a downloadable incident summary PDF.

3.6 Cybercrime Knowledge Base

Attribute	Details
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What it does	A searchable, categorized library of articles covering cybercrime types, prevention best practices, legal rights, FAQs, and glossaries. Content is authored and reviewed by security analysts.
Why it is useful	Empowers users with proactive education, reducing victimization rates and improving report quality when incidents do occur.
Data required	Article content, categories/tags, reading level (public/technical), publish date, related crime types.
Expected outputs	Search results, related article recommendations, printable guides, embedded video tutorials.

3.7 Risk Prediction System

Attribute	Details
What it does	Analyzes aggregate incident data, regional trends, and threat intelligence feeds to produce risk scores for geographic areas, industry sectors, and individual users.
Why it is useful	Enables proactive rather than reactive security posture. Authorities can pre-position resources and issue targeted alerts before spikes become crises.
Data required	Historical incident data (anonymized), incident frequency, loss amounts, crime type distribution, time-series patterns, external threat feeds.
Expected outputs	Risk score (Low / Medium / High / Critical) per region/sector, 30-day trend forecast, contributing risk factor breakdown, automated alert trigger when threshold exceeded.

3.8 Cyber Threat Alerts

Attribute	Details
What it does	Push notification system (in-app, email, SMS) distributing threat advisories to relevant users based on their profile, location, and subscribed categories.
Why it is useful	Real-time awareness enables users to take preventive action before they become victims of emerging attack waves.
Data required	Alert content, severity level, affected regions/sectors, target user segment, expiry date.
Expected outputs	Multi-channel notifications, alert banner on dashboard, alert archive, authority-published public advisories.

3.9 Data Analytics Dashboard

Attribute	Details
What it does	Role-specific analytics views. Public users see their own case statistics. Officers see jurisdiction-level metrics. Analysts and admins see platform-wide data.
Why it is useful	Turns raw report data into actionable intelligence, enabling evidence-based decision making by authorities and resource allocation by organizations.
Data required	Aggregate incident counts, financial loss totals, crime type distribution, geographic distribution, time-series data, case resolution rates.
Expected outputs	Bar charts, line graphs, pie charts, KPI cards (total incidents, open cases, avg. resolution time), exportable PDF/CSV reports.

3.10 Cybercrime Heatmap

Attribute	Details
What it does	Interactive map visualizing incident density by district, city, or state. Color-coded by risk level (green > yellow > orange > red). Filterable by crime type and date range.
Why it is useful	Immediate visual identification of crime hotspots aids law enforcement resource deployment and public awareness.
Data required	Geocoded incident location (district/city level — not exact address for privacy), crime type, incident date, case status.
Expected outputs	Interactive choropleth/heat overlay map, drill-down on region click, trend sparklines per hotspot, exportable map snapshot.

3.11 Report Escalation to Authorities

Attribute	Details
What it does	One-click escalation workflow that packages incident data, evidence, AI analysis, and guidance history into a structured authority report and forwards it to the appropriate agency (local police, national cybercrime portal, financial regulator, etc.).
Why it is useful	Removes friction from the reporting-to-authority pipeline, dramatically increasing actual reporting rates among victims.
Data required	Complete case record, evidence files with hashes, AI classification, victim consent acknowledgment, target authority endpoint.
Expected outputs	Escalation confirmation with reference number, authority-formatted PDF report, email notification to victim, webhook/API call to authority system (where integration exists).

4. Application Pages Structure

4.1 Landing Page

Aspect	Details	User Actions
Purpose	Marketing entry point; builds trust and drives registration.	Sign up, Log in, Explore features
Components	Hero with tagline, animated stats counter (total reports handled, threats identified), feature highlights, testimonials, CTA buttons.	Navigate to registration or login
Data displayed	Platform statistics (public aggregate), featured knowledge articles, latest threat alert banner.	Read alerts, browse articles

4.2 Registration & Login

Aspect	Details	User Actions
Purpose	Secure account creation and authentication.	Register, Login, Reset password
Components	Multi-step registration form (personal info, role selection, email verification), social OAuth options, 2FA setup wizard.	Submit forms, verify email, configure MFA
Data displayed	Form validation feedback, password strength indicator, 2FA QR code.	Complete onboarding steps

4.3 User Dashboard

Aspect	Details	User Actions
Purpose	Central hub showing personalized overview and quick actions.	Navigate to any feature, start new report
Components	Open cases widget, recent activity feed, active threat alerts, risk score card (for businesses), quick-action buttons.	Click through to case details, dismiss alerts
Data displayed	Own case statuses, unread notifications, recommended knowledge articles.	Manage personal data

4.4 Incident Reporting Page

Aspect	Details	User Actions
Purpose	Guided, adaptive form to capture a new cybercrime incident.	Fill and submit incident report
Components	Progress stepper, adaptive question blocks, crime-type selector with icons, date/time picker, loss amount field, optional suspect info section.	Save draft, submit, attach evidence
Data displayed	Live AI crime-type suggestion as user types description, relevant knowledge article sidebar.	Edit, preview, submit

4.5 Evidence Upload Page

Aspect	Details	User Actions
Purpose	Dedicated interface for submitting digital evidence linked to a case.	Upload, annotate, and manage evidence
Components	Drag-and-drop upload zone, file type/size validator, annotation text field per file, evidence inventory list with hash display.	Upload files, add notes, delete evidence
Data displayed	Uploaded file thumbnails, SHA-256 hash, upload timestamp, virus-scan status badge.	View, annotate, delete

4.6 Case Tracking Page

Aspect	Details	User Actions
Purpose	Real-time visibility into the status and history of a specific case.	Monitor case progress, add comments
Components	Vertical timeline of status events, assigned officer card (if applicable), AI analysis summary panel, evidence list, communication thread.	Add supplemental info, download PDF
Data displayed	Status history with timestamps, AI classification result, guidance actions taken, authority escalation reference.	Follow up, request escalation

4.7 Guidance & Recommendation Page

Aspect	Details	User Actions
Purpose	Displays the AI-generated personalized action plan for a specific incident.	Review and complete recommended steps
Components	Numbered action checklist with checkboxes, external resource links, legal information accordion, authority contact directory, escalation CTA button.	Mark steps complete, access resources
Data displayed	Prioritized actions, links to official portals, relevant laws, estimated timelines.	Export as PDF, share via email

4.8 Cyber Threat Map

Aspect	Details	User Actions
Purpose	Visual intelligence tool showing geographic distribution of cybercrime.	Explore regional risk data
Components	Interactive map (Mapbox/Leaflet), filter panel (date, crime type, severity), legend, drill-down info panel, trend sparklines.	Filter, zoom, click regions, export snapshot
Data displayed	Incident density heatmap, risk level color coding, top crime types per region, recent alert markers.	Navigate, filter, drill down

4.9 Analytics Dashboard (Authorities & Analysts)

Aspect	Details	User Actions
Purpose	Comprehensive intelligence dashboard for law enforcement and analysts.	Analyze trends, export reports
Components	KPI summary cards, bar/line/pie charts, time-range selector, jurisdiction filter, case queue table, risk prediction panel.	Filter data, schedule exports, set alert thresholds
Data displayed	Incident volumes, financial loss totals, crime type distribution, resolution rates, risk scores, prediction forecasts.	Export PDF/CSV, share dashboards

4.10 Admin Panel

Aspect	Details	User Actions
Purpose	System administration and governance interface.	Manage users, roles, and system settings
Components	User management table, role assignment modal, audit log viewer, system health monitors (CPU/memory/storage), configuration editor, data retention settings.	CRUD users, view logs, configure system
Data displayed	All registered users, audit events, system metrics, active sessions, storage utilization.	Approve officers, suspend accounts, configure AI

4.11 Knowledge Center

Aspect	Details	User Actions
Purpose	Educational library of cybercrime prevention and response content.	Search and read educational content
Components	Search bar with filters (category, reading level, date), article grid/list view, featured article carousel, video tutorial section.	Search, save articles, share, download guides
Data displayed	Article titles, summaries, categories, reading time, author (analyst/admin), publish date.	Browse, bookmark, rate content

4.12 User Profile Page

Aspect	Details	User Actions
Purpose	Personal account management and notification preferences.	Update profile and settings
Components	Personal info form, password change, 2FA management, alert subscription settings, case history summary, saved articles list.	Edit profile, manage 2FA, set notification preferences
Data displayed	Account details, security settings, notification preferences, activity summary.	Save changes, download data export, delete account

5. Functional Workflows

5.1 User Reporting a Cybercrime

1	User Navigates to Report Page Authenticated user clicks 'Report a Cybercrime' from their dashboard or the landing page CTA.
2	Select Crime Category User selects from pre-defined categories (Phishing, Fraud, Ransomware, Identity Theft, Stalking, Other). AI begins pre-loading relevant question templates.
3	Complete Adaptive Form Progressive form collects: date/time, description (with live AI suggestion overlay), financial loss (if applicable), affected platforms/accounts, suspect information (optional).
4	Review & Submit User reviews auto-generated summary. AI classification shown for transparency. User confirms consent statement and submits.
5	System Creates Case Record Backend assigns a unique Case ID, timestamps the record, persists to database, and triggers the AI Analysis pipeline asynchronously.
6	Confirmation & Next Steps User receives email confirmation with Case ID and is redirected to the Guidance page showing immediate action steps.

5.2 Evidence Submission Workflow

1	Access Evidence Upload From case detail page, user clicks 'Add Evidence'. System confirms case is in an editable status.
2	File Selection & Validation User drags files or clicks to browse. Client-side checks: file type whitelist, 50MB per file limit, total 500MB per case limit.

3**Server-Side Processing**

File uploaded via pre-signed S3 URL (direct to storage). Server: runs ClamAV antivirus scan, strips EXIF/metadata, computes SHA-256 hash, stores encrypted.

4**AI Metadata Extraction**

Background job runs OCR on images, extracts email headers from .eml files, scans documents for PII indicators. Extracted metadata appended to case record.

5**Evidence Card Created**

Evidence record persisted with: file ID, case ID, uploader ID, hash, scan status, upload timestamp. Card displayed in case evidence panel.

5.3 AI Analysis Pipeline

1**Trigger**

Case submission triggers an asynchronous AI analysis job via message queue (e.g., SQS / RabbitMQ).

2**Text Classification**

NLP model processes incident description. Outputs: crime type, sub-type, extracted entities (URLs, phone numbers, email addresses, bank account numbers).

3**Severity Scoring**

Rule-based + ML hybrid scores the incident: financial loss amount, crime type severity weight, recurrence factor, evidence strength multiplier. Score: 1-10.

4**Guidance Generation**

Template engine maps crime type + severity + victim profile to a prioritized action plan. Dynamic fields populated (e.g., relevant authority contacts for user's jurisdiction).

5**Result Storage & Notification**

Analysis results written to database. Case status updated to 'AI Analysis Complete'. User notified via in-app + email notification with link to Guidance page.

5.4 Authority Reviewing a Report

1**Case Appears in Officer Queue**

	New cases matching officer's jurisdiction appear in their 'New Cases' queue, sorted by AI severity score (highest first).
2	Officer Reviews Case Officer opens case: reads AI summary, reviews evidence (downloads with full audit log entry), reads victim guidance actions taken.
3	Officer Actions Officer can: add internal notes, request additional info from victim, re-classify crime type, assign to investigator, escalate to higher authority, or close/resolve case.
4	Status Update & Victim Notification Any status change triggers automated victim notification with a non-identifying update message (e.g., 'Your case is now under investigation').

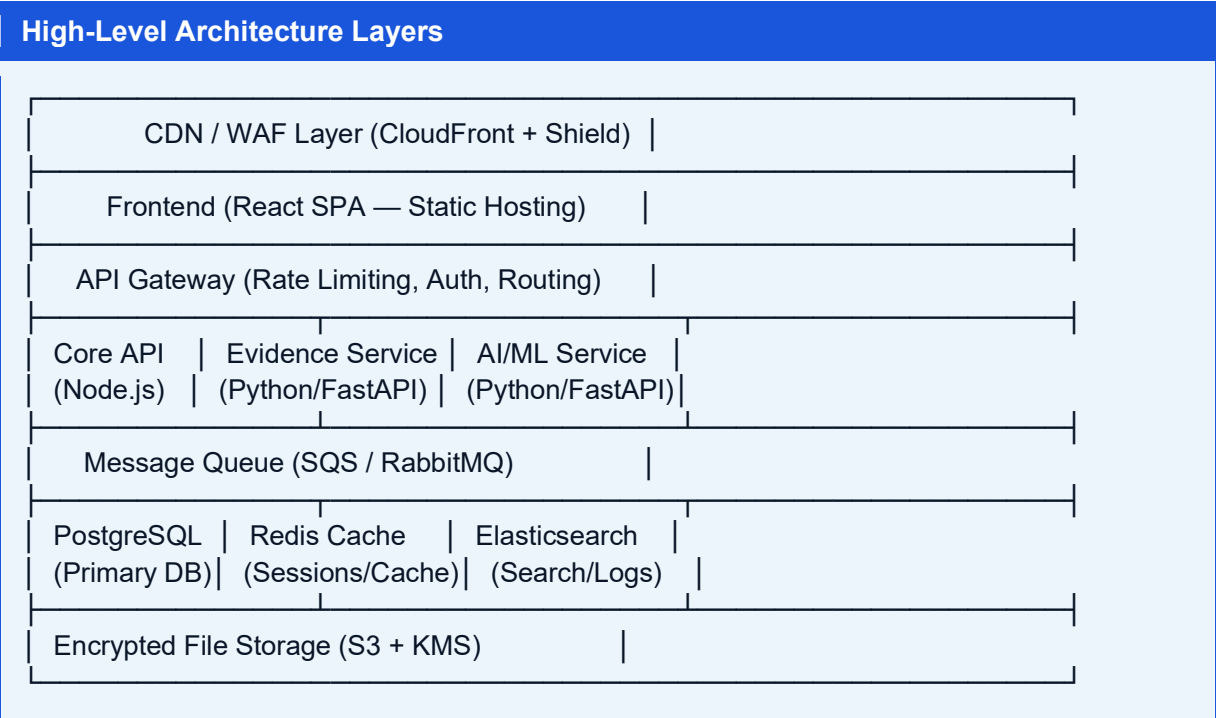
5.5 Risk Prediction Process

1	Scheduled Data Aggregation Nightly batch job aggregates anonymized incident data by region, crime type, time period, and industry sector.
2	Feature Engineering Computed features: incident frequency delta (7d/30d), loss amount trend, new crime type emergence rate, repeat victim rate, seasonal patterns.
3	Model Inference Time-series forecasting model (ARIMA / Prophet) generates 30-day incident volume forecast per region/sector.
4	Risk Score Computation Multi-factor risk score computed: forecast trend weight + current volume weight + severity composition weight. Classified: Low / Medium / High / Critical.
5	Alert Trigger If risk score crosses predefined threshold or increases by >20% week-over-week, automated threat alert is generated and published to relevant user segments.

6. System Architecture

CyberShield Nexus employs a cloud-native, microservices-inspired architecture optimized for security, scalability, and the specific requirements of sensitive evidence handling.

6.1 Architecture Overview



6.2 Component Interactions

Component	Responsibility & Interactions
Frontend (React)	Single-page application served from CDN. Communicates exclusively via HTTPS REST API calls to the API Gateway. No direct database access.
API Gateway (Kong/AWS API GW)	Enforces authentication (JWT validation), rate limiting, request routing to microservices, SSL termination, and request logging.
Core API (Node.js/Express)	Handles all CRUD operations for users, cases, guidance, alerts, and knowledge base. Publishes events to message queue for async processing.
Evidence Service (Python/FastAPI)	Manages all evidence file operations: generates pre-signed upload URLs, triggers AV scans, computes hashes, calls AI metadata extraction.

AI/ML Service (Python/FastAPI)	Consumes case-analysis jobs from message queue. Runs NLP classification, severity scoring, and guidance generation. Writes results to database via Core API.
Message Queue (SQS)	Decouples synchronous API from slow AI processing. Ensures no case analysis is lost under high load. Dead-letter queue captures failed jobs for retry.
PostgreSQL	Primary relational database. Stores users, cases, evidence metadata, guidance, alerts, and all relational data with full ACID guarantees.
Redis	Session storage, JWT blocklist (logout/revocation), short-term caching of frequently read data (knowledge articles, alert lists), rate-limit counters.
Elasticsearch	Full-text search across cases and knowledge articles. Centralized log aggregation and query. Powers the real-time search on the Knowledge Center.
S3 + KMS	Object storage for encrypted evidence files. Each file encrypted with a case-specific KMS key. Pre-signed URLs grant time-limited, scoped access.

7. Suggested Technology Stack

Layer	Technology	Rationale
Frontend	React 18 + TypeScript	Component-driven architecture, strong TypeScript typing prevents runtime errors in complex forms, large ecosystem.
UI Framework	Tailwind CSS + shadcn/ui	Rapid development, consistent design system, accessible components out of the box.
State Management	Zustand + React Query	Zustand for lightweight global state; React Query for server-state with caching, background refetching, and optimistic updates.
Maps	Mapbox GL JS / Leaflet.js	Mapbox for premium heatmap and choropleth rendering; Leaflet as a free alternative for self-hosted deployments.
Charts	Recharts + D3.js	Recharts for standard analytics charts; D3 for custom network graph visualizations (fraud network analysis).
Backend Core API	Node.js + Express / Fastify	High I/O throughput for concurrent API requests, large npm ecosystem, shared TypeScript types with frontend.
Backend AI Service	Python 3.11 + FastAPI	Python dominates the ML ecosystem; FastAPI provides async performance with automatic OpenAPI docs.
Primary Database	PostgreSQL 15	ACID compliance critical for evidence chain-of-custody integrity; JSONB columns for flexible AI metadata; mature full-text search.
Cache / Sessions	Redis 7	Sub-millisecond session lookups, TTL-based JWT blocklist, efficient rate-limit counters, pub/sub for real-time notifications.
Search	Elasticsearch 8	Powerful full-text search with relevance scoring, geospatial queries for map filtering, aggregation pipelines for analytics.
Message Queue	AWS SQS / RabbitMQ	SQS for managed cloud deployments; RabbitMQ for on-premises/sovereign data requirements. Both support DLQ for failed job handling.
File Storage	AWS S3 + KMS	Infinitely scalable; server-side AES-256 encryption; pre-signed URLs eliminate proxying large files through API servers.
NLP Classification	spaCy + HuggingFace Transformers	spaCy for fast entity extraction (URLs, emails, phone numbers); fine-tuned BERT/DistilBERT for cybercrime category classification.

Risk Prediction	Facebook Prophet / scikit-learn	Prophet handles time-series forecasting with seasonality; scikit-learn for feature-based risk scoring ensemble models.
Authentication	Auth0 / Keycloak (self-hosted)	Auth0 for managed deployments; Keycloak for sovereign/on-premises requirements. Both support OIDC, SAML, MFA, RBAC.
Email / SMS	AWS SES + SNS / Twilio	SES for high-volume transactional emails; Twilio for SMS OTP and threat alert SMS distribution.
Infrastructure (IaC)	Terraform + AWS ECS Fargate	Fargate eliminates server management; Terraform enables reproducible, auditable infrastructure.
CI/CD	GitHub Actions + AWS CodeDeploy	Automated testing, security scanning (SAST/DAST), and zero-downtime blue-green deployments.
Monitoring	Prometheus + Grafana + Sentry	Prometheus for metrics, Grafana for dashboards, Sentry for real-time error tracking across all services.
Antivirus (Evidence)	ClamAV (open-source)	On-upload virus scanning for all evidence files before storage; open-source enables self-hosted compliance.

8. Data Model

8.1 Core Entities

Users
id (UUID PK) email (unique) password_hash full_name phone role (ENUM: citizen, officer, analyst, admin) is_verified (bool) jurisdiction_id (FK, nullable — for officers) mfa_enabled (bool) created_at updated_at last_login_at is_active (bool)
Cases (Incident Reports)
id (UUID PK) case_number (sequential, human-readable) reporter_id (FK → Users) crime_type (ENUM) crime_subtype (VARCHAR) title description (TEXT) incident_date financial_loss (DECIMAL) currency affected_assets (JSONB) suspect_info (JSONB, encrypted) status (ENUM: submitted/reviewing/analysis_complete/assigned/investigating/resolved/closed) severity_score (SMALLINT 1-10) ai_confidence (DECIMAL 0-1) assigned_officer_id (FK → Users, nullable) jurisdiction_id (FK) is_escalated (bool) escalation_ref created_at updated_at
Evidence Files
id (UUID PK) case_id (FK → Cases) uploaded_by (FK → Users) original_filename storage_key (S3 object key) file_type (ENUM) file_size_bytes sha256_hash is_virus_clean (bool) scan_completed_at extracted_metadata (JSONB — OCR text, email headers, etc.) annotation (TEXT, optional) encryption_key_id (KMS key ARN) uploaded_at is_deleted (bool, soft delete)
Threat Analysis (AI Results)
id (UUID PK) case_id (FK → Cases, unique) model_version (VARCHAR) crime_type_predicted crime_subtype_predicted confidence_score (DECIMAL) severity_score (SMALLINT) extracted_entities (JSONB: {urls[], emails[], phones[], iocs[]}) severity_factors (JSONB: {financial_weight, recurrence_factor, evidence_strength}) guidance_template_id (FK → GuidanceTemplates) generated_guidance (JSONB) processed_at processing_duration_ms

Alerts
id (UUID PK) created_by (FK → Users — analyst/admin) title body (TEXT) severity (ENUM: info/warning/high/critical) crime_type (nullable) target_region_ids (INT[]) target_user_roles (ENUM[]) target_sectors (VARCHAR[]) published_at expires_at is_active (bool) view_count click_count

Risk Scores
id (UUID PK) region_id (FK → Regions) sector (VARCHAR, nullable) score_date (DATE) risk_level (ENUM: low/medium/high/critical) numeric_score (DECIMAL 0-100) incident_count_7d incident_count_30d top_crime_types (JSONB) forecast_30d (JSONB: {date, predicted_count}) model_version computed_at

8.2 Key Relationships

Relationship	Description
Users → Cases (1:N)	One user can submit many cases. Each case has exactly one reporter.
Cases → Evidence (1:N)	One case can have many evidence files. Each file belongs to exactly one case.
Cases → ThreatAnalysis (1:1)	Each case generates exactly one AI threat analysis record.
Cases → Users (Officer Assignment, 1:N)	One officer can be assigned many cases; each case has at most one assigned officer.
Alerts → Regions (N:M)	One alert can target multiple regions. Alerts table stores region_id array.
Cases → Jurisdictions (N:1)	Many cases belong to one jurisdiction. Enables officer queue filtering.
RiskScores → Regions (N:1)	Daily risk score snapshots per region enable historical trend queries.

9. AI Assistance System

9.1 Overview

The AI Assistance System is the intelligence core of CyberShield Nexus. It operates as an asynchronous microservice, consuming case data after submission and producing structured analysis results. The system combines rule-based logic for deterministic business rules with ML models for pattern recognition.

9.2 NLP Classification Pipeline

1	Text Preprocessing Incident description is normalized: lowercased, punctuation cleaned, stop words removed. Entity extraction (spaCy) identifies URLs, email addresses, phone numbers, IBAN/account numbers, and IP addresses.
2	Crime Type Classification Fine-tuned DistilBERT model (trained on labeled cybercrime incident dataset) classifies text into: Phishing, Financial Fraud, Ransomware, Data Breach, Identity Theft, Online Harassment, Sextortion, Cyber Espionage, Hacking/Unauthorized Access, Other. Output: class label + probability score per class.
3	Sub-Type Classification Second-level rule-based classifier (decision tree) maps primary type + extracted entities to sub-types (e.g., Phishing → Spear Phishing / Business Email Compromise / Smishing / Vishing).
4	Entity Enrichment Extracted URLs checked against known phishing domain blocklists (Google Safe Browsing API). IP addresses checked against abuse databases (AbuseIPDB). Results annotated on the analysis record.

9.3 Severity Scoring Logic

Factor	Scoring Logic
Financial Loss	0 loss = 1pt. <\$1K = 2pts. \$1K-\$10K = 4pts. \$10K-\$100K = 6pts. >\$100K = 8pts. >\$1M = 10pts.
Crime Type Weight	Harassment = 3. Phishing = 5. Financial Fraud = 6. Data Breach = 7. Ransomware = 8. Espionage = 9.
Evidence Strength	No evidence = 0.8x multiplier. 1-2 files = 1.0x. 3+ files = 1.2x. Video/log evidence = 1.3x.

Recurrence Factor	First report of type in region: 1.0x. Similar cases in 30 days in region: 1.1x. Active cluster detected: 1.25x.
Victim Type	Individual = 1.0x. SMB = 1.1x. Enterprise = 1.2x. Critical infrastructure = 1.5x.

Final Score = MIN(10, (LossScore × TypeWeight × EvidenceMultiplier × RecurrenceMultiplier × VictimMultiplier) / NormalizationFactor)

9.4 Guidance Generation

Guidance is generated by a template engine that maps {crime_type + severity_band + victim_type + jurisdiction} to a parameterized action plan. Templates are created and maintained by security analysts. Dynamic fields (authority contact numbers, specific legal clauses, relevant reporting portal URLs) are populated at runtime from the system database.

- Immediate actions (0-24 hours): Containment and evidence preservation steps.
- Short-term actions (1-7 days): Formal reporting, account hardening, legal consultation.
- Long-term actions (7+ days): Forensic investigation, credit monitoring, security posture review.

9.5 Model Training & Maintenance

Aspect	Approach
Initial Training Data	Synthetic dataset of 50,000+ labeled cybercrime incident descriptions generated with domain expert review. Augmented with public cybercrime report datasets.
Continuous Learning	Officer corrections (re-classifications) are collected as human-labeled samples and batch-retrained monthly.
Model Versioning	All model versions stored with performance metrics. Analysis records reference model version for auditability. Rollback capability maintained.
Explainability	SHAP values computed for severity scores, surfaced as 'Why this score?' explanations visible to officers. Builds trust in AI outputs.

10. Security Considerations

10.1 Authentication & Authorization

- Multi-factor authentication (TOTP / SMS) mandatory for officer, analyst, and admin roles; optional but encouraged for citizens.
- JWT-based stateless authentication with 15-minute access tokens and 7-day rotating refresh tokens. JWT blocklist maintained in Redis for immediate revocation on logout or compromise.
- OAuth 2.0 / OpenID Connect for social login options. No password stored for OAuth users.
- RBAC enforced at both API Gateway level (route-level permission gates) and service level (row-level security in PostgreSQL via RLS policies).

10.2 Data Encryption

Data Type	Encryption Approach
Evidence files at rest	AES-256 server-side encryption via AWS KMS. Each case uses a unique KMS data key. Key rotation enforced annually.
Sensitive fields (suspect info, victim PII)	Application-level field encryption using AWS KMS before writing to database. Encrypted at column level.
Data in transit	TLS 1.3 enforced on all connections. HSTS headers. Certificate pinning for mobile apps.
Database backups	Encrypted with separate KMS key. Stored in geographically separate region.
Session data (Redis)	Redis encryption at rest enabled. TLS in transit. Redis AUTH password + VPC isolation.

10.3 Secure File Upload

- Pre-signed S3 URLs: files upload directly from browser to S3, never passing through application servers. Upload URL expires in 5 minutes.
- File type validation: client-side whitelist (UX) + server-side MIME type inspection + magic byte validation (prevents polyglot files).
- File size limits enforced at S3 policy level (cannot be bypassed client-side).
- ClamAV antivirus scan triggered by S3 event notification immediately after upload. File quarantined automatically if threat detected.
- EXIF and metadata stripped from images before AI processing to prevent inadvertent geolocation data exposure.

10.4 Privacy Protection

- Geographic data coarsened to district/city level for heatmap display. Exact victim addresses never exposed on public-facing maps.
- Suspect information fields encrypted at rest and visible only to officers with case assignment.
- Data minimization principle: only data necessary for case analysis and authority reporting collected.
- GDPR/PDPA-compliant data subject rights: users can request full data export or account deletion (with case anonymization rather than deletion to maintain investigation integrity).
- Differential privacy noise added to aggregate statistics to prevent re-identification from small-population regions.

10.5 Infrastructure Security

- All services deployed in private VPC subnets. No direct public internet access to databases or internal services.
- WAF (AWS Shield + AWS WAF) in front of CloudFront CDN. Rules block common OWASP Top 10 attacks, SQL injection, XSS, and rate-abuse.
- Secrets Manager (AWS Secrets Manager / HashiCorp Vault) for all credentials and API keys. No hardcoded secrets in code or environment variables.
- Container images scanned for vulnerabilities before deployment (Snyk / Trivy). Base images regularly updated.
- Penetration testing conducted quarterly by external security firm. Bug bounty program for responsible disclosure.

11. Future Advanced Features

Feature	Description
AI Phishing URL Detection	Real-time URL scanning using ML model trained on phishing domain patterns. Browser extension that scans links before click. Integration into the Knowledge Center as a 'Check This Link' tool.
Scam Pattern Clustering	Unsupervised clustering (DBSCAN / k-means) on incident embeddings to automatically identify scam campaigns. When 10+ cases share similar MO, a 'Scam Campaign Alert' is auto-generated with a campaign brief for officers.
National Cybercrime Portal Integration	Webhook and API integrations with national portals (e.g., India NCRP, UK Action Fraud, US IC3). One-click escalation pre-fills the national portal form and submits on behalf of the victim.
Real-Time Threat Intelligence Feed	Integration with MISP (Malware Information Sharing Platform), VirusTotal, and Shodan APIs to enrich case analysis with external IOC (Indicator of Compromise) context.
Fraud Network Graph Analysis	Graph database (Neo4j) storing relationships between suspected fraudsters, phone numbers, bank accounts, and email addresses mentioned across cases. Network visualization reveals connected fraud rings.
Dark Web Monitoring	Integration with commercial dark web monitoring services to alert users when their credentials or data appear in paste sites or breach databases.
AI Chatbot Assistant	Conversational AI (LLM-powered) guiding users through the reporting process in natural language, available in multiple regional languages.
Mobile Application	React Native mobile app with offline-first architecture enabling evidence photo capture with GPS-tagged, timestamped metadata, and draft report creation without internet.
Deepfake Detection	Integration of deepfake detection model to analyze uploaded video evidence for AI-generated manipulation indicators.
Federated Learning for Shared Intelligence	Privacy-preserving federated learning allowing multiple national instances of CyberShield to share improved ML models without sharing raw incident data.

12. Minimum Viable Product (MVP)

The MVP focuses on the core value loop: a user files a report, receives AI-generated guidance, and an officer can review the case. All advanced analytics, prediction, and integrations are deferred to post-MVP phases.

12.1 MVP Feature Set

Feature	MVP Scope	Deferred to Post-MVP
Incident Reporting	Full adaptive form with 5 crime types, text description, date/time, loss amount	Dynamic jurisdictional routing, sub-type classification
User Authentication	Email/password + email verification + basic 2FA (TOTP)	Social OAuth, hardware key support
Evidence Upload	Image and PDF upload, SHA-256 hash, virus scan	Video upload, OCR extraction, .eml parsing
AI Classification	Crime type classification (NLP) + severity score	Entity enrichment, external blocklist checks
Guidance Engine	Static template-based action plans per crime type	Dynamic jurisdiction-specific contacts, LLM-generated plans
Case Tracking	Status timeline visible to reporter	Push notifications, SMS alerts, officer SLA tracking
Officer Dashboard	Case queue sorted by severity, basic case view	Bulk actions, advanced filters, analytics
Knowledge Base	10 foundational articles per crime type, search	Video content, analyst authoring workflow
User Roles	Citizen + Officer + Admin	Analyst role, enterprise licensing
Notifications	Email only	SMS, in-app push notifications
Analytics	Admin-only: total cases, crime type breakdown (simple charts)	Heatmap, risk prediction, exports

12.2 MVP Success Criteria

1. A citizen can successfully submit a complete incident report in under 5 minutes.
2. AI classification produces a crime type with >80% accuracy on test cases.
3. AI guidance generates a relevant, actionable checklist for the classified crime type.
4. An officer can log in, view their case queue, and update a case status.
5. Evidence upload completes with virus scan and hash verification within 30 seconds.
6. System handles 100 concurrent users without performance degradation.

13. Development Roadmap

Phase 1 — Foundation (Weeks 1-4)

Sprint 1-2: Infrastructure & Auth

Set up cloud infrastructure (VPC, ECS, RDS, S3, Redis) via Terraform

Configure CI/CD pipeline (GitHub Actions → ECS Fargate)

Implement authentication service (registration, login, email verification, JWT, 2FA)

Design and migrate PostgreSQL schema (Users, Cases, Evidence, Jurisdictions)

Build base React project structure, routing, auth context, protected routes

Sprint 3-4: Core Incident Reporting

Build adaptive incident reporting form (5 crime types)

Implement Core API CRUD for Cases

Integrate evidence upload (pre-signed S3 URLs, ClamAV, SHA-256)

Build basic case tracking page (status timeline)

Implement email notification service (AWS SES)

Phase 2 — AI Intelligence (Weeks 5-8)

Sprint 5-6: AI Classification Engine

Set up Python/FastAPI AI microservice and SQS message queue

Train and deploy DistilBERT crime type classifier

Implement severity scoring engine (rule-based + ML hybrid)

Write results to database, trigger case status update

Build guidance template engine (5 crime types x 3 severity bands)

Sprint 7-8: Officer Tools & Admin Panel

Build officer dashboard (case queue, case detail view, status updates)

Implement RBAC enforcement across all API routes

Build admin panel (user management, role assignment, audit log viewer)

Implement basic analytics for admin (case counts, crime type chart)

Accessibility audit and remediation (WCAG 2.1 AA)

Phase 3 — Intelligence Layer (Weeks 9-14)

Sprint 9-10: Analytics & Heatmap

- Build full analytics dashboard (bar/line/pie charts, KPI cards, date filters)
- Integrate Elasticsearch for case and knowledge article search
- Build cybercrime heatmap (Mapbox choropleth, crime type filters, drill-down)
- Implement data export (PDF report generation, CSV download)

Sprint 11-12: Risk Prediction & Alerts

- Implement nightly batch aggregation job
- Train and deploy Prophet time-series risk forecasting model
- Build risk score computation and storage pipeline
- Implement alert management system (create, publish, target, expire)
- Build multi-channel notification delivery (in-app, email, SMS via Twilio)

Sprint 13-14: Knowledge Base & User Experience Polish

- Build full Knowledge Center (CMS-backed articles, search, categories)
- Implement authority escalation workflow (report packaging, webhook dispatch)
- Performance optimization (query optimization, CDN caching, lazy loading)
- Comprehensive security audit (SAST, DAST, dependency scan, pen test prep)
- Beta testing with 50 pilot users (citizens + 5 law enforcement officers)

Phase 4 — Hardening & Launch (Weeks 15-18)**Sprint 15-16: Quality Assurance**

- Full regression test suite (unit + integration + E2E with Playwright)
- Load testing (k6) — target: 1,000 concurrent users, p95 response <500ms
- External penetration testing and remediation of findings
- GDPR/PDPA compliance review and documentation
- Disaster recovery drill (RTO < 4 hours, RPO < 1 hour)

Sprint 17-18: Launch Preparation

- Production environment validation and monitoring setup (Prometheus + Grafana + Sentry)
- Runbook and incident response playbook documentation
- User onboarding flow and help center content creation
- Staged rollout: internal alpha → pilot organizations → public launch
- Post-launch monitoring period with on-call team rotation

Phase 5 — Advanced Features (Months 5-12)

- Month 5-6: Mobile app (React Native), dark web monitoring integration.
- Month 7-8: Scam pattern clustering, fraud network graph (Neo4j).
- Month 9-10: National cybercrime portal API integrations.
- Month 11-12: AI chatbot assistant, multilingual support (i18n).

Milestone	Target Date
MVP Complete & Internal Alpha	Week 8
Full Feature Beta (Pilot Launch)	Week 14
Security Audit Complete	Week 16
Public Launch v1.0	Week 18
Mobile App Launch	Month 6
v2.0 (Advanced Intelligence)	Month 12

